

Runwise Developer Guide

Stack

Prerequisites

- Node.js 22 or later
 - npm
-

Setup

```
git clone https://github.com/alanwaddington/runwise.git
cd runwise
npm install
npm run dev
```

Open http://localhost:5173.

Cleaning up extraneous WASM fallback packages (optional)

npm install/npm ci always installs 5 extra packages — @emnapi/core, @emnapi/runtime, @emnapi/wasi-threads, @napi-rs/wasm-runtime, @tybys/wasm-util — even though npm ls reports them as extraneous and neither npm ci nor npm prune remove them. This is a known upstream npm limitation: they're the WASM32-WASI fallback runtime bundled with @tailwindcss/oxide and @rollup/plugin-binding (Vite's bundler), and npm doesn't correctly skip them via its cpu platform gating when they're nested this way. They're devDependencies-only, unreferenced anywhere in source, and have zero footprint in the deployed Vercel build — safe to ignore.

If you'd like to reclaim the ~9.5MB anyway:

```
npm run clean:wasm
```

This is optional, local-only, and does not change package.json or the lockfile — you'll need to re-run it after every fresh npm install/npm ci if you want to keep node_modules lean.

Project Structure

```
src/
% % % app.css           # Global styles + Tailwind theme tokens
% % % app.html          # SvelteKit HTML shell
% % % lib/
% % % % affiliates.ts   # Affiliate product definitions per route (Amazon Associates,
Garmin, and direct/non-affiliate links)
% % % % seo.ts          # SEO metadata map (PAGES), sitemap config, OG images
% % % % components/     # Shared UI components
% % % % % AdUnit.svelte # Consent-gated Google AdSense ad unit
% % % % % AdUnit.test.ts
% % % % % AffiliateLinks.svelte # Per-route affiliate product cards (Amazon, Garmin,
or direct links)
% % % % % AffiliateLinks.test.ts
% % % % % CollapsibleField.svelte # Generic animated show/hide wrapper (max-height/
opacity, aria-hidden, inert)
% % % % % CollapsibleField.test.ts
% % % % % CookieBanner.svelte # GDPR cookie consent banner (fixed bottom)
% % % % % CookieBanner.test.ts
% % % % % HeroSection.svelte
% % % % % HeroSection.test.ts
```

```

% % % % InputField.svelte
% % % % InputField.test.ts
% % % % ResultDisplay.svelte
% % % % ResultDisplay.test.ts
% % % % SeoHead.svelte # Per-page meta tags, OG, JSON-LD, AdSense account
verification
% % % % SeoHead.test.ts
% % % % SiteFooter.svelte # Footer with Privacy Policy link and Manage Cookies
button
% % % % SiteFooter.test.ts
% % % % SiteNav.svelte
% % % % SiteNav.test.ts
% % % % ToolCard.svelte
% % % % ToolCard.test.ts
% % % % ToolLayout.svelte
% % % % ToolLayout.test.ts
% % % % stores/ # Svelte stores for cross-component state
% % % % consent.ts # GDPR consent read/write (localStorage)
% % % % consent.test.ts
% % % % consentBannerVisible.ts # Writable store: true = show banner
% % % % utils/ # Pure utility modules (no Svelte dependency)
% % % % pace.ts # Pace/speed conversion functions
% % % % pace.test.ts
% % % % race-predictor.ts # Riegel formula, time parsing/formatting, prediction
table
% % % % race-predictor.test.ts
% % % % training-paces.ts # VDOT calculation (Daniels' formula), training zone
pace derivation
% % % % training-paces.test.ts
% % % % hr-zones.ts # Max HR zones, Friel LTHR zones, LTHR sub-zones,
Tanaka age estimate
% % % % hr-zones.test.ts
% % % % vo2max.ts # ACSM normative data, getFitnessCategory,
getAcsmTable, CATEGORY_COLOURS
% % % % vo2max.test.ts
% % % % parkrun.ts # Reference distance list, Riegel prediction, split
generation, PB comparison, WMA age grading
% % % % parkrun.test.ts
% % % % routes/
% % % % +layout.svelte # Root layout — CookieBanner + header + main + SiteFooter
% % % % +page.svelte # Home page — HeroSection + ToolCard grid
% % % % +error.svelte # Error page
% % % % pace/
% % % % race-predictor/
% % % % training-paces/
% % % % hr-zones/
% % % % vo2max/
% % % % parkrun/
% % % % privacy/ # Privacy Policy page

```

Design System

Tokens (`src/app.css`)

Dark mode is applied automatically via prefers-color-scheme, with an explicit `html.light/html.dark` class override (set pre-paint by `app.html`) taking priority once JS has run. Always use design tokens (`text-ink`, `text-muted`, `bg-bg`, `border-ink/10`) rather than hardcoded Tailwind colours where tokens exist.

Namespace note: color tokens must live under the `--color-*` prefix, not `--text-*` — Tailwind v4 reserves `--text-*` for font-size scale tokens (`--text-sm`, `--text-lg`, etc.), and a `--text-muted-style` color token silently compiles to an invalid font-size declaration instead of color, with no build error. (This was a real, previously-shipped bug — see PR #71.)

Text Contrast (WCAG AA)

Secondary text (help hints, labels, table headers, footer) uses `.text-muted` (!' `--color-muted`), which achieves ~4.63:1 contrast in light mode and ~6.92:1 in dark mode against the page background — both pass WCAG 2.1 AA (4.5:1 minimum). The dark-mode value is set once, at the token level, in `src/app.css` (`html.dark` and the no-JS prefers-color-scheme fallback) — never override it per-component.

An ESLint rule (runwise/no-low-contrast-text, in eslint-plugin-runwise/rules/no-low-contrast-text.js) errors on text-gray-400/text-gray-500 in Svelte files, including dark:/hover:-prefixed variants — there is no exemption for dark:text-gray-400 (it predates --color-muted's dark-mode fix and is no longer needed anywhere in the codebase).

Focus Rings (WCAG AA — SC 2.4.7)

All interactive elements (<button> and <a>) must have visible focus indicators for keyboard and assistive-technology users (WCAG 2.1 SC 2.4.7 Focus Visible).

Standard pattern:

```
<!-- Buttons -->
<button class="... focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-
accent focus-visible:ring-offset-2">

<!-- Inline links (inside text) – add rounded-sm for clean ring shape -->
<a href="..." class="rounded-sm ... focus-visible:outline-none focus-visible:ring-2
focus-visible:ring-accent focus-visible:ring-offset-2">
```

Ring offset note: Most elements use ring-offset-2. Tab buttons inside tight tablist containers use ring-offset-1 — choose whichever fits the layout.

Enforcement: The custom ESLint plugin eslint-plugin-runwise (in eslint-plugin-runwise/) provides two rules for Svelte files, both at 'error' severity in eslint.config.js:

```
eslint-plugin-runwise/
% % % index.js           # Plugin entry point
% % % rules/
% % %   require-focus-visible.js  # Checks <button>/<a> for the full focus-visible
pattern
% % %   no-low-contrast-text.js   # Bans text-gray-400/text-gray-500 (any variant) in
class attributes
```

Both rules operate on the Svelte template AST directly (SvelteElement, SvelteAttribute, SvelteLiteral) rather than plain ESTree nodes — Svelte class attributes are not standard Literal nodes, and the element name lives on SvelteElement.name.name with its attributes on SvelteElement.startTag.attributes, not on SvelteStartTag directly. Getting this AST path wrong makes a rule fail completely silently (it simply never matches, with zero lint errors either way) rather than throwing — if you write or modify a rule targeting Svelte templates, verify it actually fires against a deliberately-broken scratch .svelte file before trusting a clean npm run lint run.

Collapsible Content

Any field or block that toggles visibility based on user input (e.g. a "Custom" option revealing an extra input) should use the shared CollapsibleField component rather than a bespoke inline pattern:

```
<CollapsibleField expanded={isCustom}>
  "Ä-ç Wdf-VÆB ååå öå
</CollapsibleField>
```

Why not {#if} or the native hidden attribute:

- {#if} unmounts/remounts the content, so there's nothing to animate — the field would snap in/out instantly.
- The native `hidden` attribute also snaps instantly and sets no `aria-hidden`, so assistive technology gets no signal that the field is (or isn't) currently relevant.

What CollapsibleField does instead: the content stays mounted at all times (so its state — e.g. a partially-typed value — survives being hidden and re-shown) and visibility is purely a CSS transition (max-h-0 opacity-0 !" max-h-24 opacity-100 mb-4, overflow-hidden transition-all duration-200), paired with two accessibility attributes toggled together:

aria-hidden alone is not enough: it only affects the accessibility tree — it does not remove a nested focusable element from the tab order. Without inert, a sighted keyboard user can Tab into an invisible field (confirmed as a real, fixed bug — see PR #72 review, finding "aria-hidden properly set"). Always pair aria-hidden with inert (or tabindex="-1" on every focusable descendant, which doesn't scale) whenever hiding content that contains interactive elements.

inert's focus-blocking enforcement is real-browser behaviour that jsdom does not implement (it only reflects the IDL property, not the enforcement) — CollapsibleField.test.ts asserts the property, but the actual behavioural guarantee is covered by e2e/collapsible-field-focus.test.ts (Playwright/Chromium).

Hover Feedback (touch and mouse)

hover: classes must use the `--color-hover` token (`.hover\:text-hover`) rather than reusing `.text-ink` directly — this keeps the hover-state color relationship an explicit, named design decision instead of an incidental side effect of `--color-ink` flipping per theme.

Tailwind v4 wraps hover: utilities in `@media (hover: hover)` by default, to prevent "sticky hover" on touch devices (where a tap has no true hover-then-release gesture). This project overrides that default via a custom variant in `src/app.css`:

```
@custom-variant hover (&:hover);
```

This makes every hover: utility sitewide apply on tap as well as mouse hover — chosen deliberately so that touch users get the same hover feedback as mouse users. The accepted trade-off: tapping an element leaves it visually "stuck" in its hover-colored state until another hoverable element is tapped. If you ever need the touch-safe default back for a specific element, use an explicit `@media (hover: hover)` wrapper on that element's own styles rather than reverting the global variant.

Components

Adding a New Tool

- Create `src/routes/<tool-name>/+page.svelte`
- Import `ToolLayout` and pass `title`, `description`, and `route` props
- Add the tool to the `tools` array in `src/routes/+page.svelte`
- Add the tool link to `SiteNav.svelte`
- Register the route in `src/lib/seo.ts` (`PAGES` map) with a title, description, OG image path, and sitemap priority — this drives the page's meta tags, `sitemap.xml` entry, `robots.txt` allow rules, and JSON-LD structured data
- Add tests alongside the page

Testing

Unit and component tests (Vitest)

```
npm run test          # Run all tests once
npm run test -- --watch # Watch mode
```

Tests live alongside their source files (`*.test.ts`). Follow the existing pattern:

```
import { describe, it, expect, afterEach } from 'vitest';
import { render, cleanup, screen } from '@testing-library/svelte';
import MyComponent from './MyComponent.svelte';

afterEach(() => { cleanup(); });

describe('MyComponent', () => {
  it('renders correctly', () => {
    render(MyComponent, { props: { ... } });
    expect(screen.getByRole('...')).toBeInTheDocument();
  });
});
```

E2E tests (Playwright)

```
npx playwright install chromium # First time only
npm run test:e2e                # Run E2E tests against the production build
```

E2E tests live in the `e2e/` directory. They run against the production preview server (`npm run build && npm run preview` on port 4173). Configuration is in `playwright.config.ts`.

Code Quality

```
npm run check      # TypeScript / svelte-check
npm run lint       # ESLint
npm run format     # Prettier (auto-fix)
npm run test:e2e   # Playwright E2E tests (requires built app)
```

Workflow

This project follows an issue-driven workflow:

```
/analyse <issue>    !' requirements + acceptance criteria
/design <issue>     !' architecture + work breakdown
/develop <issue>    !' TDD implementation
/verify <PR>        !' runtime verification
/pr-reviewer <PR>  !' acceptance criteria audit
/merge <PR>         !' merge + close issues
```